CRICKET PLAYER PERFORMANCE PREDICTION USING MACHINE LEARNING

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ABSTRACT:

This project presents a machine learning approach to predicting individual player performance in the Indian Premier League (IPL). Two regression models were developed: one to predict the number of runs scored by a batsman in a match and another to predict the number of wickets taken by a bowler.

Historical IPL ball-by-ball data and match-level metadata were collected and processed to construct structured match-level datasets. Domain-specific contextual features such as recent form, venue performance, opponent statistics, career averages, strike rate, and economy rate were engineered. XGBoost regression models were trained using chronological train-test splitting to prevent data leakage.

Model performance was evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). The final model achieved strong predictive performance. SHAP analysis was used to interpret feature contributions and ensure model transparency. The system was deployed using an interactive Streamlit dashboard for real-time prediction and visualization.

INTRODUCTION:

With the growth of analytics in professional sports, data-driven decision-making has become essential in cricket. In tournaments such as the IPL, teams analyze historical performance data to optimize team selection, bowling strategies, and batting order decisions.

Predicting player performance is a complex task due to the influence of multiple contextual factors, including:

- Recent performance trends
- Venue conditions
- Opposition strength
- Long-term consistency
- Match dynamics

Traditional statistics such as career averages do not account for contextual variations. Therefore, this project aims to build a machine learning-based prediction system that integrates both historical performance and contextual information.

PROBLEM STATEMENT:

The objective of this project is to build predictive models that:

1. Estimate runs scored by a batsman in a given IPL match.
2. Estimate wickets taken by a bowler in a given IPL match.

The system must:

- Avoid data leakage
- Use contextual feature engineering
- Provide interpretable predictions
- Be deployable via an interactive interface

DATASET DESCRIPTION:

The dataset consists of IPL historical data across multiple seasons and includes two primary components:

Ball-by-Ball Dataset:

This dataset contains delivery-level information such as:

- Match ID
- Batsman
- Bowler
- Runs scored
- Extras
- Dismissal type
- Over and ball number
- Batting and bowling teams

Match Summary Dataset:

This dataset includes contextual attributes such as:

- Match date
- Venue
- Season
- Toss winner
- Match winner

Both datasets were merged using match ID to create a unified structured dataset.

DATA PREPROCESSING:

Data preprocessing involved the following steps:

Column Standardization:

All column names were converted to lowercase and standardized to remove inconsistencies.

Match ID Resolution:

Different naming conventions for match identifiers were unified before merging datasets.

Data Cleaning:

- Missing values were handled.
- Duplicate records were removed.
- Numeric columns were validated.

Legal Deliveries Identification:

Wides and no-balls were excluded when calculating balls faced and overs bowled.

Match-Level Aggregation:

Ball-by-ball data was aggregated to match-level performance:

For batsmen:

- Total runs scored
- Balls faced
- Boundaries

For bowlers:

- Total runs conceded
- Legal deliveries bowled
- Wickets taken

This created structured datasets suitable for regression modeling.

FEATURE ENGINEERING:

Feature engineering was central to improving predictive performance.

Batsman Features:

• Career Average Runs
• Venue-Specific Average Runs
• Opponent-Specific Average Runs
• 10-Match Rolling Average (Recent Form)
• Strike Rate
• Boundary Count

The rolling average was computed using shifted windows to avoid data leakage.

Bowler Features:

• Career Average Wickets
• Venue-Specific Average Wickets
• Opponent-Specific Average Wickets
• 10-Match Rolling Wicket Average
• Economy Rate

These features capture both long-term ability and short-term performance trends.

MODEL DEVELOPMENT:

XGBoost regression was selected due to its strong performance on structured tabular data and ability to model nonlinear relationships.

Train-Test Split:

A chronological 80–20 split was used:

- First 80% → Training
- Last 20% → Testing

This ensures future data is not used to predict past performance.

Training Strategy:

- Hyperparameters were tuned experimentally.
- Learning rate and depth were optimized.
- Early stopping was applied using a validation set.
- Separate models were trained for batsmen and bowlers.

MODEL EVALUATION:

Model performance was evaluated using:

• Root Mean Squared Error (RMSE)
• Mean Absolute Error (MAE)

Batsman Runs Prediction Results:

Model	RMSE	MAE
XGBoost	4.68	3.20

Interpretation:

The model prediction differs from actual runs by approximately 3 runs per match on average. Given the variability in T20 cricket, this represents strong predictive performance.

Bowler Wickets Prediction Results:

Model	RMSE	MAE
XGBoost	0.68	0.46

Interpretation:

The model prediction differs from actual wickets by less than half a wicket on average. This indicates strong model accuracy for wicket prediction.

MODEL EXPLAINABILITY (SHAP ANALYSIS):

SHAP was used to interpret model predictions.

In SHAP summary plots:

- Red dots represent high feature values
- Blue dots represent low feature values
- Points on the right increase prediction
- Points on the left decrease prediction

Key findings:

- Recent form was the most influential feature for batsmen.
- Venue performance significantly impacted predictions.
- Economy rate strongly influenced wicket predictions.

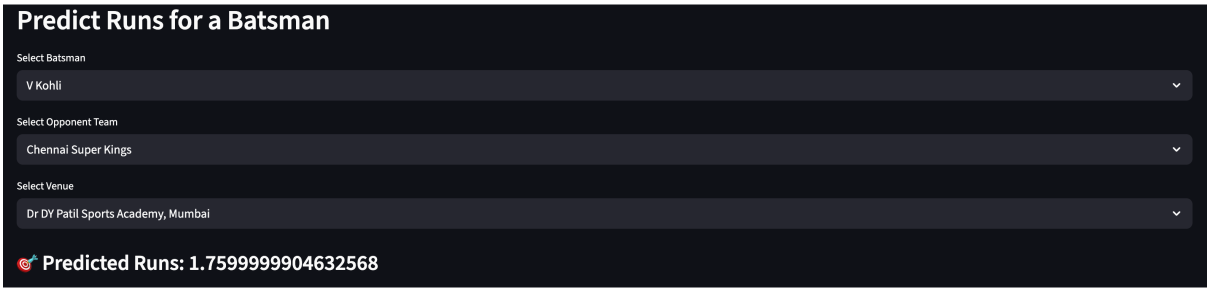
SHAP ensures transparency and helps validate that predictions align with cricket logic.

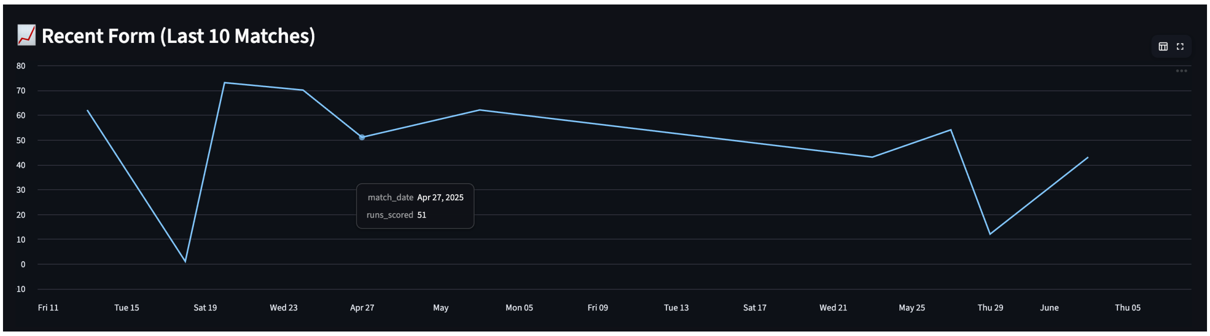
STREAMLIT DASHBOARD DEPLOYMENT:

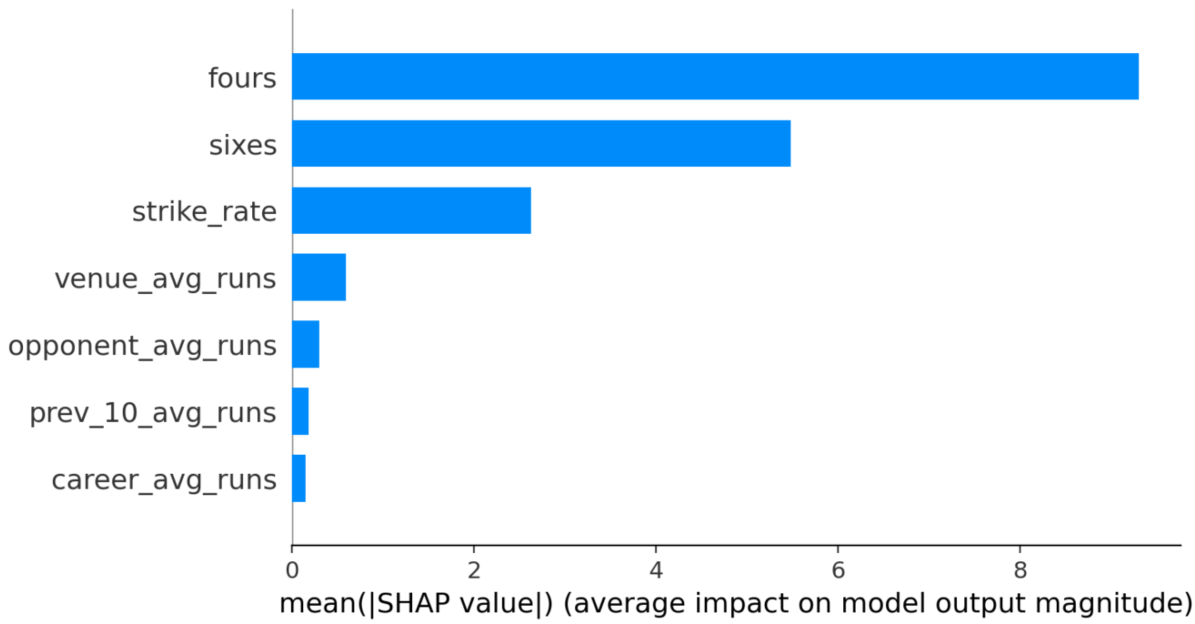
An interactive Streamlit dashboard was developed to demonstrate the system.

The dashboard allows users to:

- Select player
- Select opponent
- Select venue
- View predicted performance
- View recent form graph
- View SHAP-based feature importance







CONCLUSION:

This project demonstrates a complete end-to-end machine learning pipeline applied to sports analytics. Through structured preprocessing, domain-aware feature engineering, advanced regression modeling, and explainability techniques, accurate and interpretable IPL player performance predictions were achieved.

The deployment through Streamlit provides practical usability and demonstrates real-world applicability of machine learning in sports analytics.

FUTURE WORK:

- Incorporate pitch and weather conditions
- Include player workload and fitness data
- Extend prediction to match outcome modeling
- Deploy system on cloud infrastructure
- Explore deep learning approaches